

Usage Pattern Analysis of Smartphone Users

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Abstract—Recently, mobile traffic has been tremendously increased due to the existence and deployment of smart devices such as a smartphone. These devices are using various types of access networks such as 3G, WiFi, and mobile WiMAX. Network service providers are also selling these access networks with various types of plans. Therefore, the need for managing smart devices and mobile networks are increasing. However, research on mobile network management has been focusing on performance of network itself. Few research has focused on applying usage patterns of smartphone users to mobile network management. In this paper, we present analysis results of usage patterns to present how smartphone users use their phones. We define five possible states of a smartphone based on a phone's basic operation. We collect real usage log data from real smartphones of real users during a two month study. We show that all users have their own usage pattern which present their characteristics. We present a case study how to apply this usage pattern to power management of smartphones. We also discuss how to apply it to mobile device management and network management.

Index Terms—Usage pattern analysis, Smartphone management, Mobile network management, Personalization

I. INTRODUCTION

As the growth of mobile networks and services has been accelerated, mobile traffic generated from mobile devices has been tremendously increased [1] and mobile devices (e.g., mobile handsets or PDAs) have emerged as important tools in our lives [2]. Mobile devices are used for various applications such as making voice/video calls, browsing the Internet, playing games and so on. Furthermore, mobile devices has multiple network interfaces which can access multiple access networks such as 3G, WiFi, and mobile WiMAX. Network operators also provide new pricing systems, data plans, and value-added services due to a variety of access networks. Hence, the need of mobile network management is getting bigger for supporting the growth of mobile networks and mobile devices.

Existing mobile network management has focused on performance of network itself and requirements in the view of network operators. However, the requirements for improving performance in the view of customers and providing personalized services are growing. Some of the recent researches have monitored the perceived quality and usage patterns of customers who are using mobile devices in mobile networks and applied them to various areas [3]–[12].

In this paper, we present usage pattern analysis of smartphone users. A usage pattern means how users use their phones. First, we define possible smartphone states based on its basic functions, e.g., voice call and data communication. Second, we define log metrics for measuring time and battery spent in each operational state. Third, we present to develop a mobile application, (called a *battery logger*), for collecting the log data from real smartphones and deploy it to real users. Finally, we describe to analyze the collected data for showing that each user has his/her own usage pattern. In this research, we developed a battery logger based on an Android mobile platform [13] and collected log data from Android smartphone users. In all, we collect logs of real smartphone usage during two months from 20 users as the first step of this research. The main goal of our study is to observe the high-level usage patterns of real smartphone users in the wild, and understand the implications of these characteristics for managing smartphones, and their networks.

The remainder of this paper is organized as follows. Section II describes collecting and analyzing users' activities from smartphones as related work. In Section III, we present to collect, analyze, and apply usage patterns from smartphones. In Section IV, we present case studies of applying usage patterns to resource management of smartphones and mobile networks. Finally, conclusions are drawn and future work is discussed in Section V.

II. RELATED WORK

Recently, many researches have focused on customer's usage analysis on mobile devices.

Demumieux and Losquin [3] describes data analyzed by gathering functionalities used (directory, calendar, SMS' number received and sent), durations and navigations (presses on the keypad and history of windows opened) from Windows CE and Symbian 6 and 7 OS). Reades et al. [4] presented real-time and historical visualizations of mobile phone usage levels in central Rome for understanding urban systems. They developed and analyzed the much larger samples of existing data generated daily by mobile networks from real telecommunication operators. Rahmati and Zhong [5], [6], [10] presented to analyze human-battery interaction from mobile phones for energy-efficient design for a longer battery lifetime. They evaluated various aspect of human-battery interaction,

including charging behavior, battery indicators, user interfaces for power-saving settings, user knowledge, and user reaction which are a kind of usage patterns. Our previous work proposed a method of predicting the battery lifetime of mobile devices based on usage patterns [7]. We defined the possible state of mobile devices based on their operating functions and develop a method of predicting battery lifetime based on average battery consumption and duration of each state. However, one of the limitations is that we did not use real data from real devices. In this research, we present to analyze real data for overcoming the limitation. Oliver presented preliminary results of a large-scale smartphone user study that examines how users interact with and consume energy on their personal mobile devices [8]. This study used over one millennium of user interaction traces from over 17,300 BlackBerry users. Verkasalo defined and discussed a framework for mobile audience measurements called MobiTrack [9]. These measurements are all about data collection at the point of convergence – mobile devices – rather than network or gateway side measurements, or user surveys. Shye et al. presented a comprehensive analysis of real smartphone usage during a 6-month study, involving 25 users, and one specific smartphone, the Android G1 [11]. The goal of this work was to study the high-level characteristics of smartphone usage, and to understand the implications on optimizing smartphones, mobile embedded architecture, and their networks. This work motivated us to start this analysis study of usage patterns from smartphone users. This work is similar to our work but they were focusing on applying it to mobile architecture. We are focusing on how to characterize personalized usage patterns for mobile network management. Shepard proposed LiveLab, a methodology to measure real-world smartphone usage and wireless networks with a reprogrammable in-device logger designed for long-term user studies [12]. They presented an iPhone 3GS based deployment of LiveLab with 25 users intended for one year. They demonstrated the temporal dynamics and trends of application usage, as well as the difference between individual application and application categories. This work presented a good methodology for measuring data from networks and users, but it only focused on application usage.

In this paper, we collect low level data for analyzing usage patterns of smartphone users and show how each user's usage pattern is differentiated from other users' pattern by collecting real data from real smartphone users.

III. USAGE PATTERN ANALYSIS

In this section, we present our hypothesis of this study, a logger application, a collecting method, and analysis results.

A. Overview

In this study, our hypothesis is that we can provide personalized usage patterns by analyzing users' activity on smartphones. We also assume that each user activity on a smartphone is ~~mutual~~ independent. For validating the hypothesis, we collected the following data from smartphones.

- Voice call status (Ringing, Waiting, Calling)

- Screen status (On/Off)
- 3G data communication status (In/Out/InOut)
- Active network (3G, WiFi)
- WiFi status (On/Off)
- Battery level (0–100 %)
- Battery status (Charging, Discharging, Full)
- Battery plugged status (Battery, AC, USB)

B. Data Collection

First, we developed a log collecting mobile application based on the Android mobile platform. This application monitors the previous defined data and records it to a log file periodically. It transfer the log file to the data analysis server every midnight through 3G or WiFi determined by a user. A smartphone has many operation states, but we considered only five operation states which are a large influence on power consumption: a) voice call, b) data communication via WiFi, c) data communication via 3G, d) waiting, and e) other activity. After collecting those data periodically, we calculated the time and battery spent in each state and compared usage patterns among smartphone users.

First of all, we deployed the application on the project homepage [14] on February 17, 2011. To obtain users for the study, we announced on the online bulletin board for anonymous volunteers in our campus and posted online advertisements to social network services such as Twitter or Facebook for obtaining more users outside campus.

Overall, we collected 40 users from March to April, 2011. For this paper, we used the logs from the 20 users with the longest total recorded time. The data from these 20 users is 1.5 Gbytes and 1.26 Mbytes per each log on average.

C. Analysis Results

We present our analysis results of collected smartphone usage data.

1) *Average usage*: First, we present average usage time and battery spent in each operation states (Fig. 1). As shown in Fig. 1a, most users spent time in a waiting state (85% and 54%). Although this shows an average usage pattern, we need to consider each user's usage pattern which we will mention in Section III-C2. Fig. 1b shows a comparison of battery consumption among all operating states. It shows different rates compared to Fig. 1a because battery consumption rate of each state is different.

2) *Usage Pattern*: Fig. 2 shows a time and battery consumption comparison of individual user's smartphone usage percentage of five operational states: a) waiting, b) voice call, c) data communication via WiFi, d) data communication via 3G, and e) Etc. From the spent time comparison (Fig. 2a), time which each user spent in each state is different. The state where an user spent more time will be used more in the future based on Zipf's law [15] and each user has their own usage pattern in the long-term view. Hence, we assume that this usage pattern is one of the user's characteristics and the ultimate goal of our study is to make an analytic model of users' behavior on using a smartphone. Fig. 2b shows a

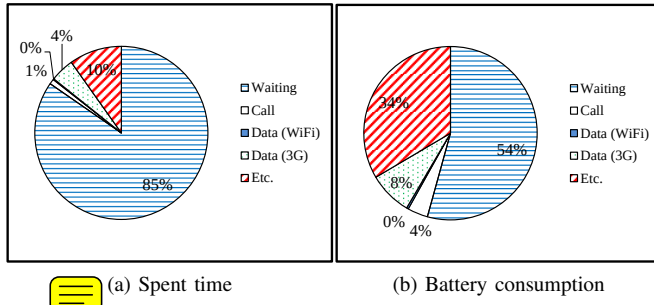


Fig. 1. Average smartphone usage percentage of five operational states in terms of (a) spent time and (b) battery consumption.

battery consumption spent in each state of individual users and differences from time pattern shown in Fig. 2a. It means that battery consumption in each state is different. As shown in Fig. 1a, most users spent more time on average in the waiting state, but each user had a different rate in all usages.

Fig. 3 shows a comparison of individual user's smartphone usage of three main operational states for communications: a) voice call, b) communication via WiFi, and c) communication via 3G. In Fig. 3a, each user has different time percentages spent in each state. In Korea and our campus, WiFi networks are deployed widely. However, most users have been using 3G as a major communication method. We present some reasons about it. First, although WiFi supports fast speed and high bandwidth, it is inconvenient because its coverage is smaller than that of 3G and it is difficult to support seamless horizontal handover. Second, this comparison is based on time spent in each state. When we transfer a same data packet, the time spent in WiFi is shorter than that of 3G because the speed of WiFi is faster than that of 3G. We will present a comparison of session duration of WiFi and 3G in Section III-C3. Third, most participants come from our campus. From March 2011, our campus has been completely covered by WiFi networks. So, most users did not notice the WiFi coverage. However, we expect that this network usage pattern will be slightly changed in the future. We expect that the changes of IT services will affect to smartphone usage patterns. Finally, WiFi is less secure than 3G. Recently, personal information in smartphones and credit card information collection via WiFi have been introduced.

We can know users' characteristics from this chart. For example, User12 is a heavy WiFi user because of spending more time in the WiFi state than that in other states. User14 is a heavy 3G user because of spending more time in the 3G state than that in other states. User18 is a heavy voice call user because of spending more time in the call state than that in other states. From this usage pattern comparison, we can guess an individual price plan and recommend a personalized price plan for optimizing smartphone usages. From Fig. 3a, we guess that User12 has a limited data plan and User14 has an unlimited data plan. Actually, User14 is the first author who is using an unlimited data plan. User12 is a friend of the first author who is using a limited data plan.

Fig. 3b shows a comparison of battery consumption of three main operational states. It shows that the battery consumption is different by a type of networks.

3) *Network Usage*: An analysis of network usages is important for modeling the networks that support smartphones. In this section, we study voice call, 3G, and WiFi characteristics for the entire logger session, including both plugged and unplugged time.

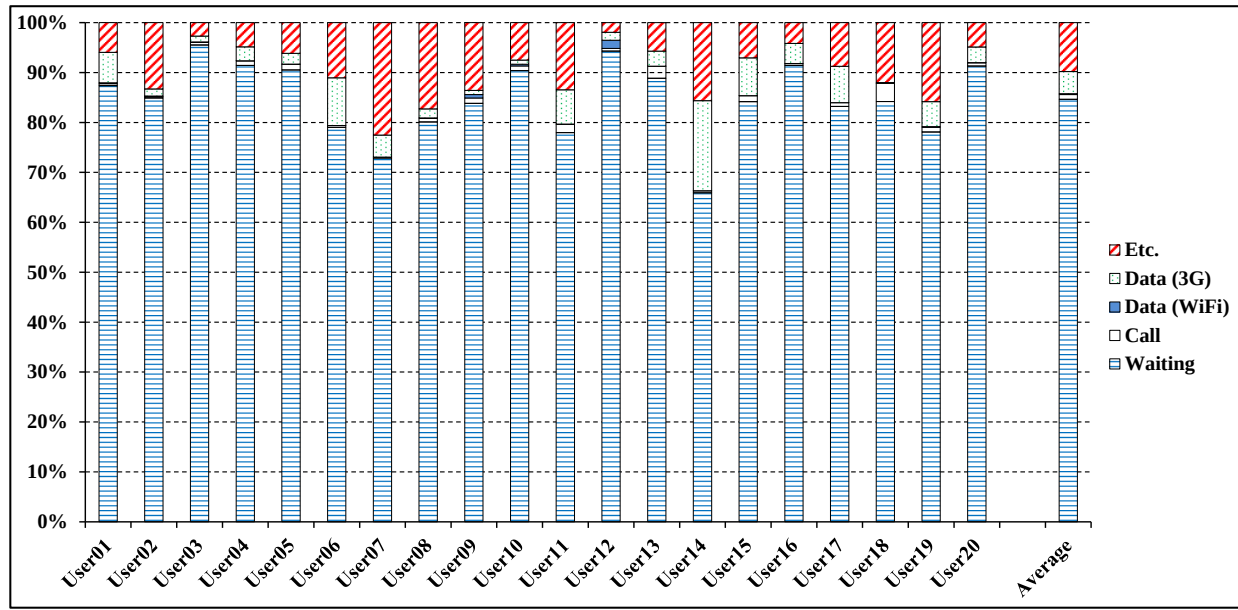
Fig. 4 shows cumulative distribution functions (CDFs) of a voice call, 3G, and WiFi network session durations for all users, User12, User14, and User18. As we mentioned in the previous section, we selected three users because their usage patterns shows clear characteristics in terms of 3G, WiFi, and voice call. We present session durations on a log scale due to readability. From the figure, we can see how much time a user spends in a session.

Fig. 4a shows a CDF for the session durations across all of users. As shown in the figure, the 3G network has many short session durations that dominate the samples, whereas the voice call has many long session durations. Because data communication via the 3G network is expensive, it has a short duration. The WiFi network has longer session durations than the 3G network.

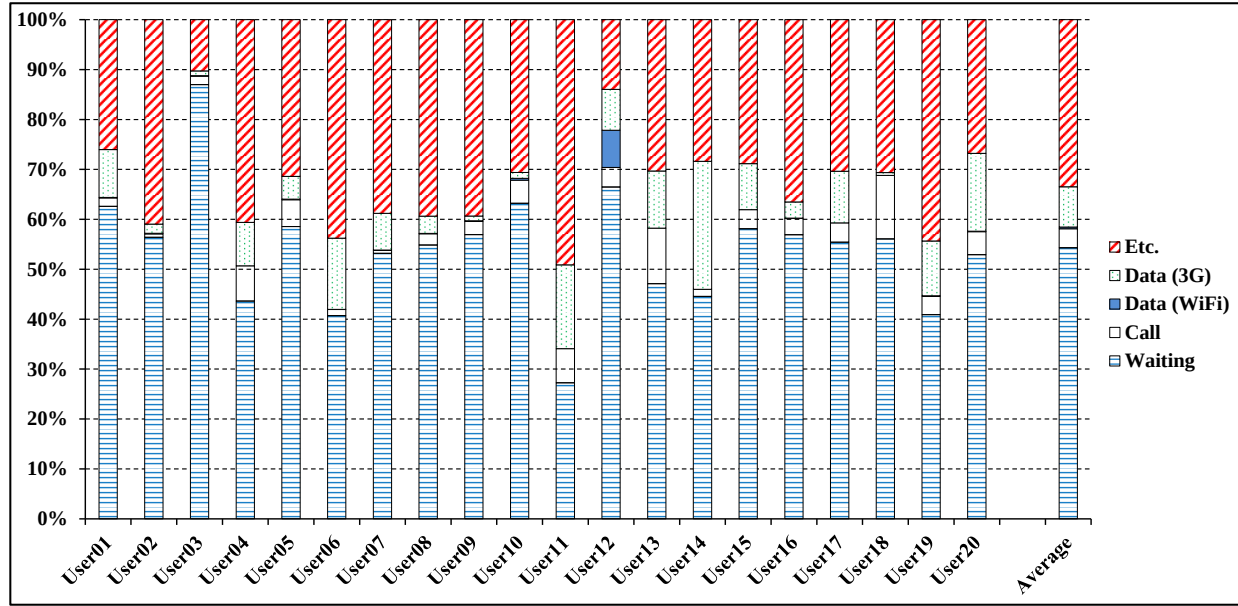
However, the session durations of all users are also different. Fig. 4b shows a CDF of session duration of User12. When this user uses the WiFi network, the session duration is longer than that of other users. That is, we can know that this user is a heavy WiFi user. Fig. 4c shows a CDF of session duration of User14. When this user uses the 3G network, the session duration is longer than that of the WiFi network. That is, we can know that this user is a heavy 3G user. Fig. 4d shows a CDF of session duration of User18. When this user uses the voice call, the session duration is longer than that of the 3G and WiFi network comparably. That is, we can know that this user is a heavy voice call user. From the differences of session durations, we can know that each user has his or her own characteristics and we can use it for optimizing communications in a smartphone and create new services.

4) *Battery Usage*: Understanding of human-battery interaction in a smartphone is important for supporting an efficient battery usage [6]. To study the battery usage of our users, we extract battery intervals from each of the logger sessions. A battery interval represents a time interval where we can account for the battery being plugged via AC and a USB cable and unplugged (battery).

Fig. 5 shows a comparison of smartphone power sources of all users: a) battery, b) USB, and c) AC. Each user has a different pattern in this case. For example, User1 connected his or her smartphone to an AC power source in proportion of 30%. We can know that User1 worked inside where the AC power source is available or kept to connect his or her smartphone to an AC power source during sleeping. User13 connected his or her smartphone to a USB power source in proportion of 12% and almost did not use an AC power source. We can know that User13 is near a computer or any device which has USB power sources. User14 connected his or her



(a) Spent time



(b) Battery consumption

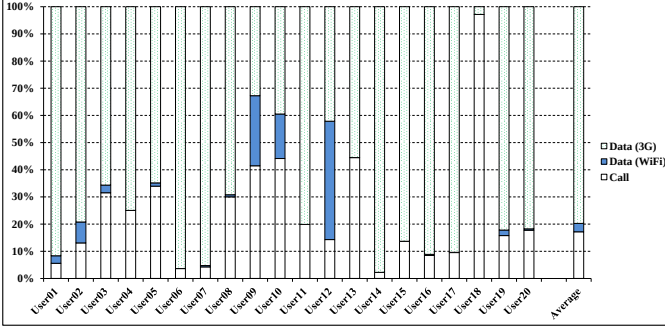
Fig. 2. A comparison of Individual user's smartphone usage percentage of five operational states in terms of (a) spent time and (b) battery consumption.

smartphone to a AC and USB power source in proportion of 28% and 22%, respectively. We can know that User13 is also working inside. User20 used his or her smartphone with a battery power source. We can know that this user always changed a battery when it run out. We will present a battery change interval, a start battery level, and an end battery level for showing details. A start battery level is an indicated level when a user starts to change a battery or charge with an AC or USB power source. An end battery level is an indicated level when a user finishes to change a battery or charge with an AC or USB power source.

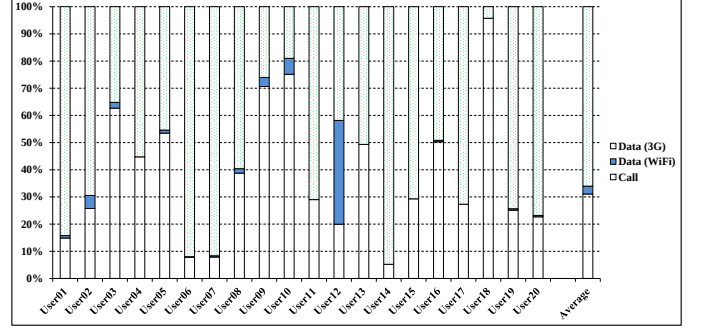
Fig. 6 shows CDFs of battery change interval of all users,

User1, User13, User14, and User20. User20 who uses his or her smartphone without connecting to an AC or USB power source changes a battery periodically and in a shorter time than average users. Whereas, User1, User13, and User14 who connected their phones to an AC and/or USB power source changes a battery in a longer time than average users.

Fig. 7 shows CDFs of battery level when smartphone users charge or change battery at a session on a start and end point. As shown in Fig. 7a, User20 changed a battery when a battery level is lower then 40%, whereas, User1, User13, and User14 changed or charged a battery frequently regardless of a battery level. We can know that User20 charge his or her phone based



(a) Spent time



(b) Battery consumption

Fig. 3. A comparison of individual user's smartphone usage percentage of three main operational states (call, WiFi, and 3G) in terms of (a) spent time and (b) battery consumption.

on charge level feedback from the battery interface and other users often charge their phone, regardless of the charge level. As shown in Fig. 7b, when a user finished to change or charge his or her smartphone, a battery level of User20 is higher than 90%. In the analysis of battery usage, we showed that users have their own characteristics. These analysis results contribute to insights into improving mobile phone design for users to effectively deal with battery and apply personalization to battery management [6].

D. Discussion

In this section, we showed that smartphone users had their own usage patterns which represents variable time and battery spent in all operating states. If we can make an analytic model for usage patterns, we can apply it to optimize a smartphone design and mobile networks in terms of energy efficiency. Actually, the WiFi network is a low price, a fast transfer speed, and high bandwidth, but our users did not use WiFi as a major communication method in their smartphone. Specially, the introduction of an unlimited data plan of 3G has affected users to utilize it as a major communication method.

IV. CASE STUDY

In this section, we provide case studies for applying smartphone usage pattern analysis.

A. Usage pattern based battery lifetime prediction

A mobile device's short battery lifetime can cause much inconvenience or reduce its usefulness ([16], [17]). Mechanisms to provide long and stable battery life are required. One of the methods to guarantee long battery life is to minimize battery consumption by reducing useless battery usage.

The key to successfully minimizing battery consumption is accurately predicting the battery consumption of mobile devices' various operating states. In our previous work [7], we proposed a method for predicting available battery lifetime in mobile devices based on the usage patterns of individual users. However, we did not validate it with a real usage pattern data from smartphone users. In this section, we present a usage pattern based battery lifetime prediction as a case study.

First of all, we assume that a mobile device has n possible states and r users are using the mobile device. We define a number of symbol descriptions and formulations as follows:

$$\vec{B} = (B_1, \dots, B_n), \quad (1)$$

$$\vec{p} = (p_1, \dots, p_n), \quad (2)$$

$$\vec{R} = (R_1, \dots, R_n), \quad (3)$$

$$R_i = p_i \cdot B_i, \quad (4)$$

where B_i is the average battery consumption rate of the i th state, and p_i time consumption rate of the i th state, which is determined to satisfy $\sum_{i=1}^n p_i = 1$. Because each user has a different vector $\vec{p}$ according to the usage pattern, this vector is defined for the symbol that represents the usage pattern. R_i is the ratio of battery consumption of the i th state.

We can formulate the given problem as follows.

Problem formulation: Determine the average battery consumption rate vector $\vec{B}$ and the usage pattern vector $\vec{p}$ to minimize the mean square error (MSE) of an predictor

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] \quad (5)$$

where $\hat{\theta}$ is an estimator, our prediction method, and θ is the true values measured from users. We predict the available battery lifetime based on usage patterns using the following formulation.

$$T = \frac{V}{\sum_{i=1}^n R_i} = \frac{V}{\sum_{i=1}^n p_i \cdot B_i}, \quad (6)$$

where T is the available time predicted by usage patterns, and V is the total battery capacity. Because each user has a different $\vec{p}$, every user's battery has a different remaining lifetime. Previous approaches cannot represent the user's personal characteristics, because they do not consider usage patterns.

We define two symbols that represent the usage patterns of each user.

$$\vec{p}_k = (p_k^1, \dots, p_k^n), \quad (7)$$

$$\vec{U} = (U_1, \dots, U_r), \quad (8)$$

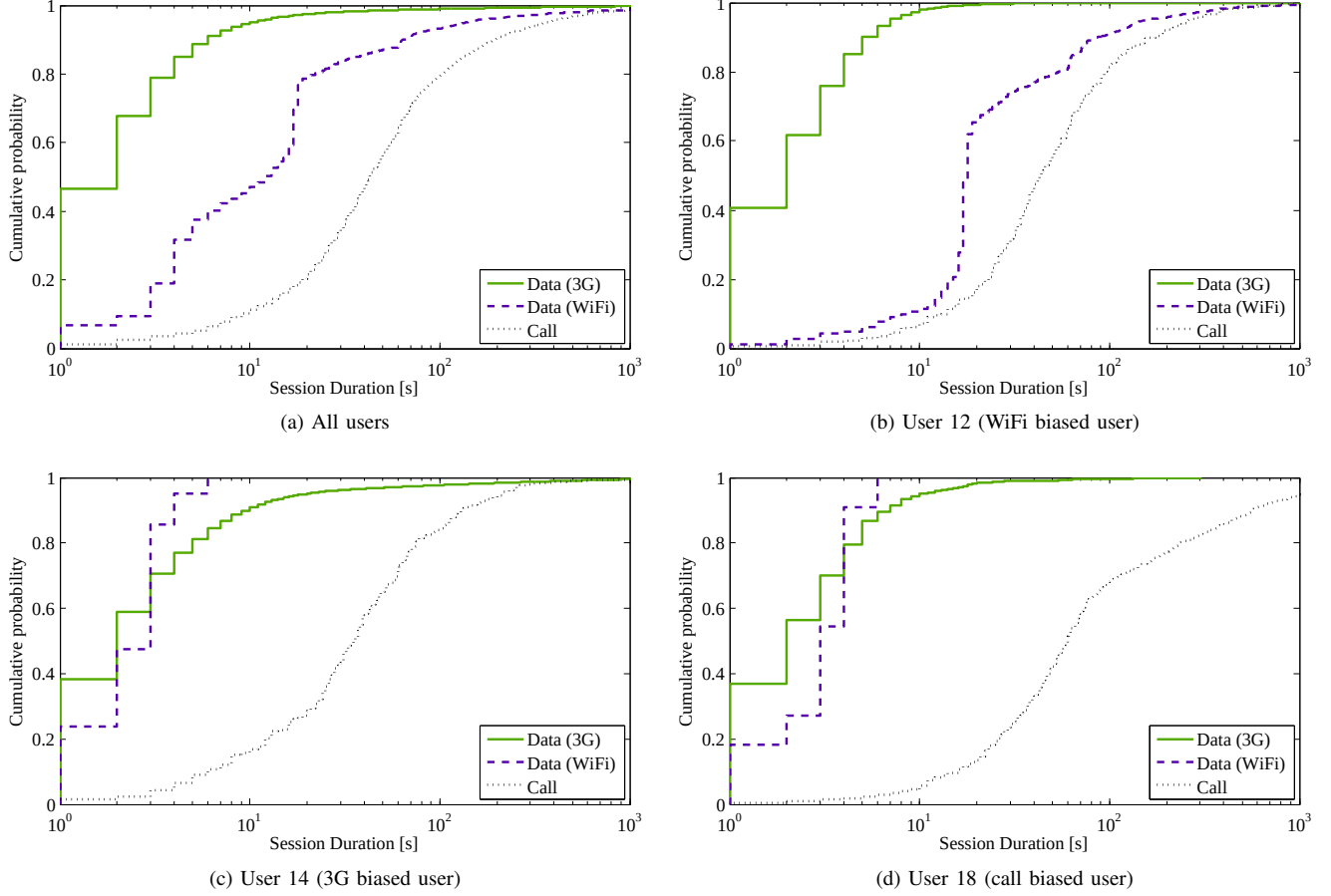


Fig. 4. CDFs of call, WiFi, and 3G session duration

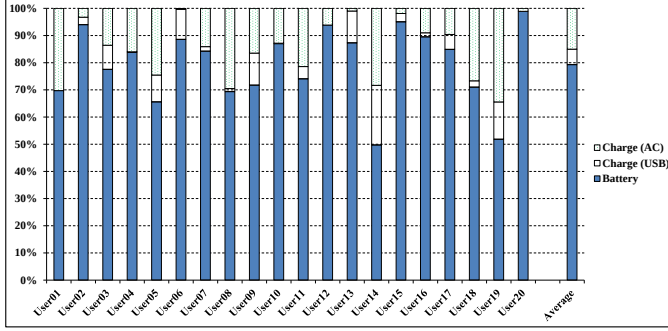


Fig. 5. A comparison of smartphone power source (battery, USB, and AC) percentage in terms of spent time.

where $\vec{p}_k$ is the usage pattern of the k th user, p_k^i is the ratio of operating time that the k th user spent in the i th state, and U_k is the average battery consumption rate of the k th user considering the usage pattern—That is, p_k^i represents the probability that the k th user will spend time in the i th state because we assume that the usage pattern will be consistent with the user's history. In this paper, we defined five possible states: a) waiting, b) voice call, c) data communication via WiFi, d) data communication via 3G, and e) other activity.

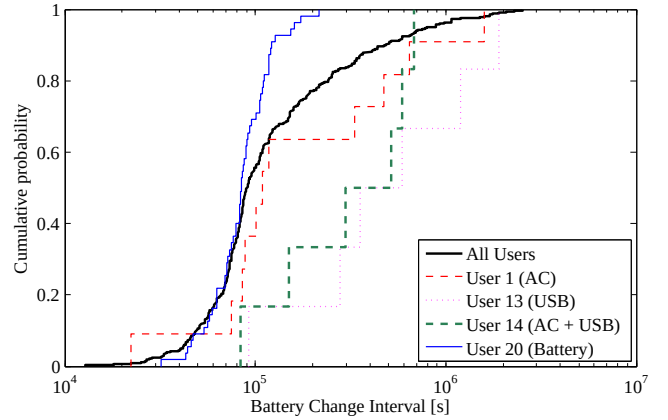


Fig. 6. CDFs of battery change interval; User1: AC biased, User13: USB biased, User14: both AC and USB biased, and User20: battery biased.

Based on these states, we present how we can predict the battery lifetime if the average battery consumption rate and usage patterns are known. The average battery consumption rates ($\vec{B}$) of each state based on the collected time-series data are as follows:

$$\vec{B} = (0.048, 0.252, 0.300, 0.132, 0.271), \quad (9)$$

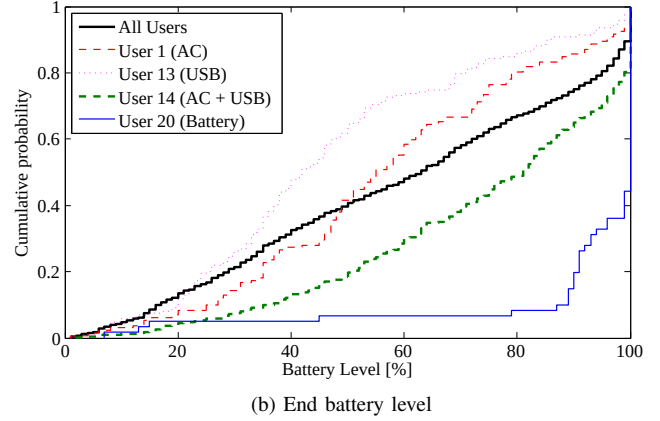
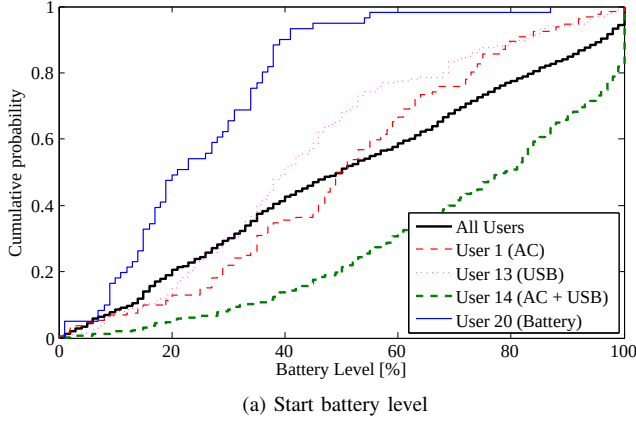


Fig. 7. CDFs of battery level when smartphone users charge or change battery at a session (a) start and (b) end point of time; User1: AC biased, User13: USB biased, User14: both AC and USB biased, and User20: battery biased.

If the usage pattern is given as Fig. 2a, we can define $\vec{p}$ for User12, User14, and User18 as follows:

$$\vec{p} = \begin{pmatrix} 0.94504 & 0.00527 & 0.01345 & 0.01622 & 0.02003 \\ 0.63739 & 0.00788 & 0 & 0.22672 & 0.12801 \\ 0.81914 & 0.04231 & 0.00001 & 0.00146 & 0.13708 \end{pmatrix},$$

where the row value represents the usage pattern of each user and the column value represents each state: column 1 is *Waiting*, column 2 is *VoiceCall*, column 3 is *WiFi*, column 4 is *3G*, and column 5 is *Other*. For example, User12 spent more time in the WiFi state than other users. User14 spent more time in the 3G state, and User18 spent more time in the voice call state. To predict the battery lifetime of users with different usage patterns, we compare our proposed method with a method based on the static value which is an arithmetic mean of all battery consumption rates.

The average battery consumption rate based on an arithmetic mean is about 0.2006. However, the average battery consumption rate for each user by our proposed method are as follows:

$$\vec{U} = \vec{p} \cdot \vec{B}^T = \begin{pmatrix} 0.058294 \\ 0.097198 \\ 0.087325 \end{pmatrix}, \quad (10)$$

where the average battery consumption rate of User1 is 0.058294, that of User2 is 0.097198, and that of User3 is 0.087325. From this case study, we can present different battery consumption rates for each user, based on his or her usage patterns. Furthermore, we can adaptively predict the battery lifetime of a mobile device using dynamic usage patterns.

B. Other Useful Cases

A usage pattern analysis can be applied to personalized access network selection in heterogeneous wireless networks [18] and the extensive research on energy-efficient design for a longer battery lifetime [10]. It can be also used to enable effective battery usage by reducing useless battery consumption and detects abnormal battery usage by comparing

operating time between normal and abnormal states. We utilize it for user identification as a security aspect.

V. CONCLUDING REMARKS

A smartphone is one of the important devices in the current mobile networks. Hence, we need to manage smartphones for mobile network management. In this paper, we presented usage pattern analysis using log data collected from smartphones. This usage pattern represents different time spent in a voice call, data communication, and waiting state for each user. Our study is the first step to monitor smartphone usages and construct usage patterns. We presented our analysis result using *real usage log data* from *real smartphones* of *real users*. Our result contributes network providers to install mobile networks based on usage patterns which are different in places and users, and to optimize network infrastructures and create new services by analyzing usage patterns. It can also be used access network selection and energy efficient communication based on usage patterns which are constructed from real data of customers [10], [18]. It contributes to optimize battery usage on smartphones, predict personalized battery lifetime, and detect abnormal processes and intrusion, and identify users.

As future work, we are continuously collecting usage log data from smartphone users for showing large-scale analysis results. We will add more data such as amount of network traffic and signal strength as well as running applications. We will also analyze users' characteristics for supporting usage patterns. Finally, we will apply our usage pattern to managing mobile networks.

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